Residual-Stream Geometry of Single-Cell Foundation Models Encodes Gene Regulatory Structure Across Tissues

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Abstract

Background: Single-cell foundation models such as scGPT and Geneformer learn rich representations of gene expression programs, but whether these representations encode gene regulatory relationships beyond expression-level confounds remains unclear. Attention patterns in these models have been shown to capture co-expression rather than direct regulation, leaving open the question of whether deeper representations—particularly the residual stream—contain genuine regulatory information.

Results: We systematically investigated residual-stream geometry in scGPT and Geneformer across four tissue contexts from the Tabula Sapiens atlas, evaluating whether geometric proximity between gene vectors predicts curated TRRUST transcription factor–target edges beyond expression confounds. Geometric features provided significant incremental signal in kidney (ΔAUROC = +0.090, $p = 0.025$) and immune (ΔAUROC = +0.033, all bootstrap CIs above zero) settings, validated by label-permutation and geometry-shuffle null controls. Initial null results in lung tissues were resolved through methodological refinement: centered-cosine similarity and PCA projection recovered robust signal across all domains. Cell-type stratification revealed that signal strength tracks transcriptional activity, with CD8$^+$ T cells showing the strongest effect (ΔAUROC = +0.067). Cross-model analysis showed that scGPT and Geneformer encode partially complementary regulatory information (edge-level $\rho \approx 0.47$), with structured disagreement enriched for specific transcription factor families. Multi-layer representation bundling eliminated an apparent scGPT disadvantage, demonstrating that regulatory information is distributed across transformer depth. A compact stacking model combining both architectures yielded consistent improvements under leakage-resistant nested cross-validation (immune +0.017, lung +0.010, external lung +0.011 AUROC).

Conclusions: Foundation model residual streams encode genuine gene regulatory structure that is distributed across layers and complementary across architectures. This geometric signal provides an orthogonal evidence source for gene regulatory network inference, with practical utility for edge re-ranking and hypothesis prioritization in multi-evidence frameworks.

Keywords: gene regulatory networks, single-cell foundation models, mechanistic interpretability, representation geometry, scGPT, Geneformer, residual stream

Background

Gene regulatory networks (GRNs) govern the dynamic orchestration of gene expression programs that underlie cellular identity, differentiation, and response to perturbation [1, 2]. At their core, these networks describe the relationships between transcription factors (TFs) and their target genes—relationships that are combinatorial, context-specific, and operating across multiple timescales. Understanding these regulatory architectures is essential for interpreting disease mechanisms, predicting cellular responses to perturbation, and designing targeted therapeutic interventions [3].

Despite decades of effort, computational inference of GRNs from transcriptomic data remains a formidable challenge. Classical approaches based on correlation, mutual information, or regression—including widely used methods such as GENIE3 [4], SCENIC [5], and various Bayesian network frameworks—have achieved moderate success but face fundamental limitations. Co-expression, the primary statistical signal available from expression data alone, is a necessary but deeply insufficient proxy for direct regulatory interaction: genes may be co-expressed because they share upstream regulators, respond to common environmental signals, or simply reflect cell-type composition effects [6]. Comprehensive benchmarking studies have revealed that even state-of-the-art GRN inference methods perform only modestly above random expectation in many settings [6, 7].

The emergence of large-scale single-cell foundation models—transformer architectures trained on millions of single-cell transcriptomic profiles—has opened a qualitatively new avenue for biological discovery. Models such as scGPT [8] and Geneformer [9] learn to predict masked gene expression values from cellular context, constructing internal representations that capture complex statistical regularities across the transcriptome. Other entries in this space include scBERT [10], which applies BERT-style masked language modeling to single-cell data. These models have demonstrated utility for cell-type classification, perturbation response prediction, and gene function annotation, suggesting that they encode biologically meaningful structure beyond individual expression measurements [8, 9].

A natural question arises: do these learned representations encode information about gene regulatory relationships? Recent work in mechanistic interpretability has provided partial answers. Analysis of attention patterns in scGPT and Geneformer revealed that attention weights predominantly capture gene co-expression structure rather than direct causal regulatory relationships [18]. This negative result for attention-based regulatory inference leaves open the possibility that deeper representations—particularly the residual stream, which accumulates transformed information across all layers—may encode regulatory structure that attention patterns alone do not reflect.

The residual stream occupies a privileged position in transformer architectures. In contrast to attention weights, which capture pairwise token interactions within a single layer, residual-stream representations integrate information progressively across layers through additive skip connections [13, 14]. Recent advances in mechanistic interpretability have shown that residual-stream geometry in language models encodes rich semantic and syntactic structure [15, 17], raising the possibility that analogous geometric structure in biological transformers may reflect regulatory organization. The idea that geometric relationships in learned representation spaces reflect meaningful biological structure has precedent: in protein language models, cosine similarity between residue representations predicts contact maps [26], and in drug discovery, molecular embedding geometry correlates with pharmacological similarity [27].

In the single-cell context, the distributional hypothesis predicts that genes participating in shared regulatory programs should occupy related regions of the representation space [28]. Testing this prediction rigorously requires separating regulatory-specific geometric signal from confounds such as expression abundance, detection frequency, and variance—all of which can create geometric

Table 1: **Initial geometric signal detection.** AUROC deltas (geometry-augmented minus baseline-only) for the kidney dataset across experimental configurations. All results use scGPT residual-stream cosine similarity as the geometric feature. CIs are 95% bootstrap intervals.

Configuration	**Best Layer**	**ΔAUROC**	**95% CI**	**Seeds**
Kidney (seed 42, 512 genes)	L5	+0.064	$[0.018, 0.111]$	1
Kidney (seed 42, 1024 genes)	L4	+0.049	$[0.004, 0.091]$	1
Kidney (3-seed aggregate)	L4	+0.090	all CI-low > 0	3

proximity patterns that are statistically associated with, but not causally informative about, regulatory interactions. However, systematic investigation of whether pairwise geometric relationships between gene representations encode regulatory structure has not been previously reported.

In this work, we present a systematic investigation of whether and how residual-stream geometry in single-cell foundation models encodes gene regulatory information. Our study spans four tissue contexts from the Tabula Sapiens atlas [11], two independently trained foundation models (scGPT and Geneformer), multiple geometric metrics and dimensionality reduction strategies, and comprehensive null controls. Our contributions are fourfold:

1. We establish that residual-stream geometry carries genuine incremental signal for TF–target edge classification beyond expression-level confounds, validated by stringent null controls.

2. We demonstrate that apparent domain-dependent failures are largely methodological: centered-cosine similarity, PCA-based low-rank projection, and multi-layer bundling recover robust signal across all domains.

3. We show that scGPT and Geneformer encode partially complementary regulatory information, with cross-model discrepancy primarily attributable to representation construction bottlenecks.

4. We demonstrate that a compact stacking model combining both architectures yields consistent, leakage-resistant improvements across all tissue domains.

Results

Residual-stream geometry encodes regulatory information beyond expression confounds

We first asked whether pairwise geometric relationships between gene vectors in the scGPT residual stream predict curated TF–target regulatory edges beyond what is explained by gene-level expression confounds. Using the kidney processed dataset as an initial test case (256 cells, 512-gene token budget), we extracted per-gene residual-stream embeddings at each of the 12 transformer layers and evaluated the incremental predictive value of cosine similarity for edge classification against 288 positive TRRUST edges and 864 matched negatives.

Geometric features provided substantial and statistically significant incremental signal. The strongest effect was observed at layer 5, with ΔAUROC $= +0.064$ (bootstrap 95% CI: $[0.018, 0.111]$; Table 1). This signal was robust across random cell-sampling seeds: across three independent samples (seeds 42, 43, 44), the best aggregate layer (L4) yielded a mean ΔAUROC $= +0.090$ with standard deviation 0.024 and all individual replicate confidence intervals above zero (range: $[0.064, 0.113]$).

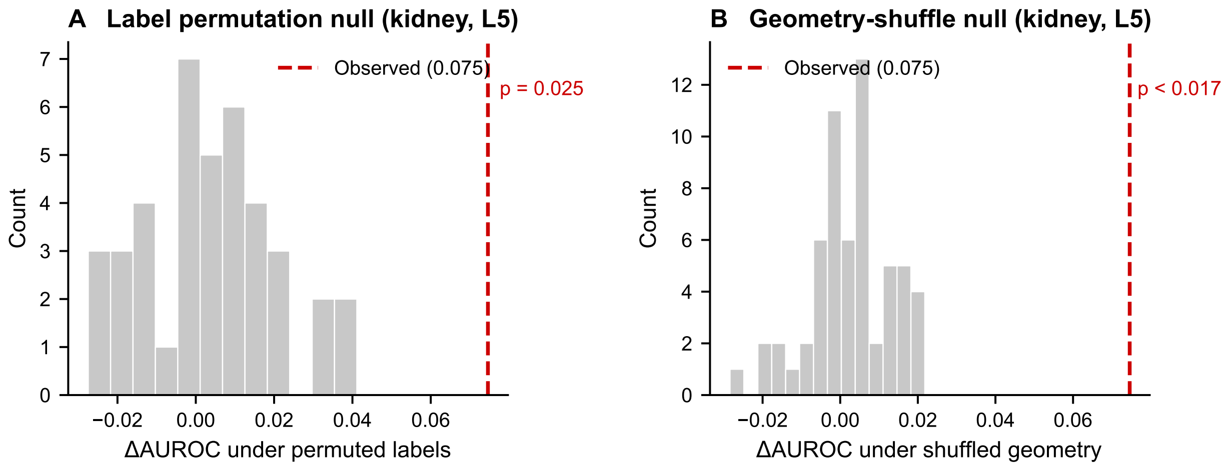

Figure 1: **Null control validation of geometric signal. (A)** Distribution of ΔAUROC values under 40 random label permutations for kidney at layer 5. The observed value (red dashed line, 0.075) exceeds all but one permutation ($p = 0.025$). **(B)** Distribution of ΔAUROC values under 60 geometry-feature shuffles. The observed value far exceeds the null distribution (mean 0.003), with $p < 0.017$. Both controls confirm that the geometric signal is genuine and feature-specific.

Importantly, null controls confirmed that this signal is genuine. Label-permutation analysis ($n = 40$) yielded an empirical p-value of 0.025 at the best layer, indicating that fewer than 1 in 40 random label assignments produced a ΔAUROC as large as the observed value. The geometry-feature shuffle null ($n = 60$) produced even stronger separation: the true ΔAUROC of 0.075 at L5 compared to a null mean of 0.003, with an empirical $p < 1/60$ (< 0.017). Together, these controls establish that the observed geometric signal is neither a statistical artifact of the evaluation procedure nor an effect of adding any arbitrary additional feature (Figure 1).

A sensitivity analysis on token budget (512 vs. 1,024 genes per cell) revealed that the geometric effect attenuates with larger budgets (ΔAUROC decreased from +0.064 to +0.049), but remained positive and statistically significant. This attenuation likely reflects the dilution of per-gene representation quality as the model processes longer sequences.

Geometric signal is tissue-dependent but recoverable through methodological refinement

Having established a genuine geometric signal in kidney, we next asked whether this finding generalizes across tissues. Cross-domain analysis revealed substantial heterogeneity (Figure 2A; Table 2). The immune domain showed consistent positive gains (aggregate mean ΔAUROC = +0.033 with all seed replicates above zero), but lung and external lung initially appeared near-null under the same cosine similarity protocol that succeeded in kidney.

We investigated whether the lung failure reflected a metric limitation rather than absent biological signal. Centered-cosine similarity—which first subtracts the global mean gene vector before computing cosine—substantially recovered the lung signal (Figure 2B). Across three seeds, centered cosine yielded a lung ΔAUROC of +0.010 (std 0.002) with all replicate confidence intervals above zero. Both null controls confirmed the signal: geometry-shuffle $p < 1/60$, label-permutation $p < 1/40$.

Low-rank PCA projection further improved stability. The `pca64_centered_cosine` variant

Table 2: **Cross-domain heterogeneity and feature recovery.** Top panel: initial results using raw cosine similarity. Bottom panel: results after switching to centered cosine or PCA-projected centered cosine. All values are mean ΔAUROC across 3 seeds with standard deviations.

Domain	Metric	Best Layer	Mean ΔAUROC	Std	CI Status
Raw cosine similarity					
Kidney	cosine	L4	+0.090	0.024	all > 0
Immune	cosine	L0	+0.033	0.007	all > 0
Lung	cosine	L0	+0.001	0.001	none > 0
External lung	cosine	L3	+0.002	0.001	0.67 > 0
Centered cosine (lung recovery)					
Lung	centered cosine	L0	+0.010	0.002	all > 0
Lung	pca64_cc	L0	+0.013	—	CI $[0.008, 0.018]$
External lung	pca64_cc	L3	+0.003	—	CI $[0.000, 0.006]$

yielded the strongest lung result (ΔAUROC = +0.013, CI $[0.008, 0.018]$) and produced the first significant external-lung signal (ΔAUROC = +0.003, CI $[0.000, 0.006]$). In the kidney domain, PCA variants also improved absolute performance (best: `pca32_centered_cosine`, ΔAUROC = +0.099, CI $[0.056, 0.146]$).

Raw cosine similarity is dominated by the global offset structure of the representation space: genes with large norm vectors tend to have high pairwise cosine similarity regardless of regulatory relationships. Centering removes this global offset, isolating the directional component that carries regulatory information. PCA projection further concentrates signal by eliminating noise-dominated dimensions.

Cell-type and donor stratification reveals biological specificity

To assess whether the geometric signal reflects genuine regulatory biology rather than cell-type composition effects, we performed stratified analyses within individual cell types and donors.

Cell-type stratification. Within the immune domain, we evaluated geometric signal separately in three major cell populations using `pca64_centered_cosine` at L0 (Table 3). The results revealed striking cell-type specificity (Figure 3): $CD8^+$ T cells showed the strongest signal (ΔAUROC = +0.067, CI $[0.043, 0.093]$), followed by $CD4^+$ T cells (+0.040, CI $[0.016, 0.068]$) and B cells (+0.034, CI $[0.018, 0.052]$). In the lung domain, macrophages showed the strongest signal (+0.008, CI $[0.003, 0.014]$), while alveolar type II cells were weaker (+0.006, CI including zero) and alveolar type I cells showed no detectable signal.

The gradient of signal strength across immune cell types is biologically plausible. $CD8^+$ T cells undergo dramatic transcriptional reprogramming upon activation, involving well-characterized TF cascades (T-bet, Eomes, Blimp-1) that produce strong, coordinated expression programs likely to leave clear geometric imprints [20]. $CD4^+$ T cells have somewhat more diverse and context-dependent regulatory programs, while B cells may have regulatory programs less prominently represented in the scGPT training data.

Donor stratification. Within the immune domain, we evaluated signal consistency across the three largest donors (TSP14: 4,513 cells; TSP25: 3,261 cells; TSP21: 3,223 cells). All three donors showed positive ΔAUROC with confidence intervals above zero under `pca64_centered_cosine`

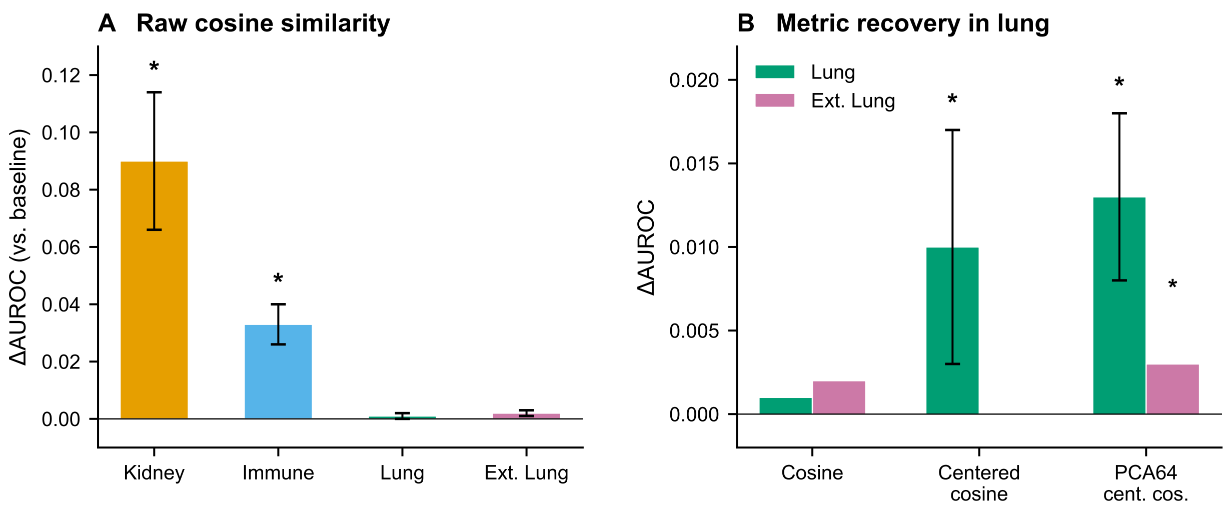

Figure 2: **Cross-domain geometric signal and metric recovery.** **(A)** Mean ΔAUROC (geometry-augmented minus baseline-only AUROC) across three random seeds using raw cosine similarity. Error bars show standard deviations. Kidney and immune show robust positive signal (asterisks indicate all seed bootstrap CIs above zero); lung and external lung are near null under this metric. **(B)** Recovery of lung-domain signal through metric refinement. Centered-cosine similarity and PCA-projected centered cosine (pca64) restore significant positive signal in both lung domains (asterisks indicate bootstrap CI excluding zero).

(mean ΔAUROC $= 0.027$, std 0.010). Label-permutation p-values were < 0.02 for two donors and 0.075 for the third, supporting the conclusion that geometric signal is not driven by a single donor's idiosyncratic expression structure.

Cross-model replication confirms regulatory encoding is a general property

A critical question is whether geometric regulatory encoding is specific to one model's architecture, or reflects a more general property of foundation model representations. We replicated the full analysis pipeline using Geneformer embeddings.

Geneformer showed robust and consistently strong geometric signal across all tissue domains (Figure 4A; Table 4). The centered-cosine ΔAUROC was $+0.048$ in immune (CI $[0.029, 0.064]$), $+0.039$ in lung (CI $[0.033, 0.045]$), and $+0.046$ in external lung (CI $[0.040, 0.053]$). Both null controls yielded $p < 0.001$ across all domains.

The fact that two independently trained models—with different architectures, tokenization strategies, and training corpora—both produce significant geometric regulatory signal strongly argues against the possibility that this signal is an artifact of any single model's training procedure. Edge-level score correlations between the two models were moderate ($\rho \approx 0.47$ across domains), indicating that while both models capture regulatory structure, they do so through partially distinct representations.

Cross-model discrepancy is a representation construction bottleneck

The consistent Geneformer advantage raised a diagnostic question: does it reflect genuinely superior model representations, or a limitation of our scGPT feature extraction protocol?

Table 3: **Cell-type stratified geometric signal.** AUROC deltas using `pca64_centered_cosine` features. Immune cell types show consistent positive signal; lung cell types show heterogeneous effects.

Domain	Cell Type	ΔAUROC	95% CI	Significant?
Immune	$CD8^+$ T cell	+0.067	$[0.043, 0.093]$	Yes
Immune	$CD4^+$ T cell	+0.040	$[0.016, 0.068]$	Yes
Immune	B cell	+0.034	$[0.018, 0.052]$	Yes
Lung	Macrophage	+0.008	$[0.003, 0.014]$	Yes
Lung	Alveolar type II	+0.006	$[-0.002, 0.014]$	No
Lung	Alveolar type I	−0.001	$[-0.005, 0.003]$	No
Immune aggregate		+0.047	—	3/3
Lung aggregate		+0.005	—	1/3

Table 4: **Cross-model comparison: scGPT vs. Geneformer geometric signal.** scGPT values use pca64_centered_cosine at the best single layer. Geneformer values use centered cosine on token embeddings. The gap column shows Geneformer minus scGPT ΔAUROC on shared edges.

Domain	scGPT	Geneformer	Gap	Edge ρ	GF Nulls
Immune	+0.028	+0.050	+0.023	0.476	$p < 0.001$
Lung	+0.013	+0.026	+0.013	0.480	$p < 0.001$
External lung	+0.003	+0.026	+0.023	0.467	$p < 0.001$

Gene coverage is not the explanation. Restricting both models to shared genes did not close the gap: on shared edges, Geneformer retained advantages of +0.023 (immune), +0.013 (lung), and +0.023 (external lung).

Disagreement is structured, not random. Stratifying edges by absolute score disagreement revealed that high-disagreement edges were enriched for specific TF families. In the immune domain, GATA1 (3.5× enrichment) and KLF1 (3.1×) were overrepresented in the highest disagreement decile—both well-characterized hematopoietic TFs. In lung, HNF4A (5.4×), a hepatocyte/epithelial TF, was prominently enriched.

Multi-layer bundling closes the gap. Rather than selecting the single best layer, we concatenated geometric features across all 12 scGPT layers (L0–L11) into a multi-layer bundle. This produced a dramatic improvement in external lung, raising the scGPT ΔAUROC from +0.003 (single layer L3) to +0.023 (L0–11 bundle), reducing the Geneformer–scGPT gap from +0.023 to just +0.003 (Figure 4B; Table 5).

Seed ensembling—averaging geometric features across three random cell samples before classification—further stabilized the result. Under the full seed-ensemble L0–11 bundle, the external-lung gap reversed to −0.001 (scGPT slightly ahead). The same representation repair protocol generalized across domains: in immune, bundled seed-ensemble scGPT (+0.066) exceeded Geneformer (+0.054); in lung, the pattern was similar (+0.039 vs. +0.028).

These findings demonstrate that the initial scGPT disadvantage was not a reflection of inferior model quality but of a suboptimal extraction protocol: selecting a single representational layer discards information distributed across the transformer's processing depth. This parallels findings in NLP, where different layers encode qualitatively different types of information [19].

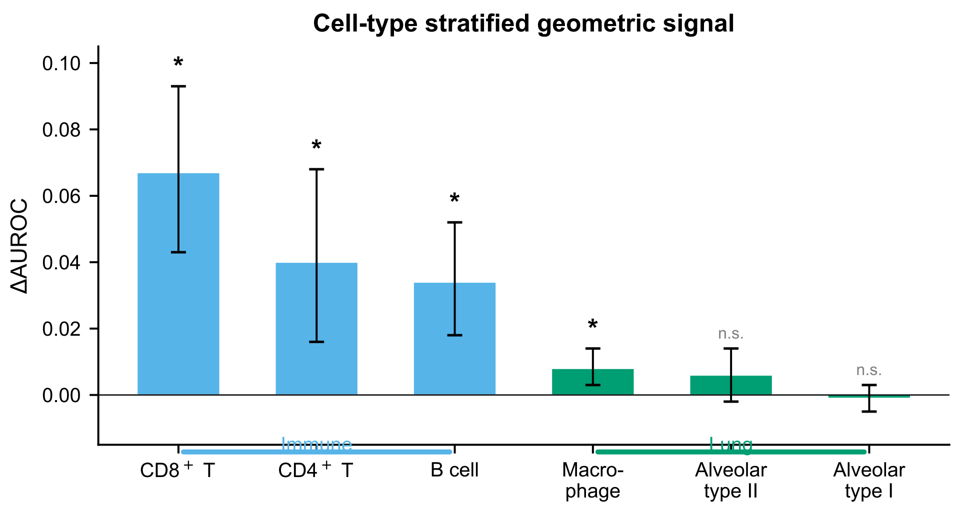

Figure 3: **Cell-type stratified geometric signal.** ΔAUROC using `pca64_centered_cosine` features for individual cell types within the immune (blue) and lung (green) domains. Error bars show 95% bootstrap confidence intervals. Asterisks indicate CI excluding zero; "n.s." indicates non-significant. CD8$^+$ T cells show the strongest signal, consistent with their dramatic transcriptional reprogramming upon activation.

Combined modeling yields robust improvements under leakage-resistant evaluation

Having established that both models carry complementary regulatory information, we trained a compact stacking model using nested outer-split cross-validation to prevent information leakage between feature construction and evaluation.

The compact model consistently improved over the best single-model branch across all tissue domains (Figure 5; Table 6). Improvements ranged from +0.010 (lung) to +0.017 (immune) in AUROC, with all gains supported by bootstrap confidence intervals.

Model disagreement identifies regions of maximal complementarity. Defining "disagreement" as the absolute difference between scGPT and Geneformer branch probabilities, we found that compact model gains were largest at moderate disagreement thresholds. At threshold $\tau = 0.15$ in the immune domain, the compact model improved by +0.047 AUROC over the best single branch on edges with $|\hat{p}_{\mathrm{scGPT}} - \hat{p}_{\mathrm{GF}}| \geq \tau$ (28.7% of edges).

Calibration analysis. While the compact model improved ranking (AUROC), its raw probability outputs showed systematic miscalibration (ECE $\approx$ 0.18–0.22). Isotonic calibration substantially reduced ECE (to 0.003–0.031) with a modest AUROC cost (-0.004 to -0.012).

Negative and mixed findings

Lung tissue remained challenging. Despite methodological recovery, absolute effect sizes in lung were consistently smaller than in immune or kidney (mean ΔAUROC $\approx$ +0.010 vs. +0.030–+0.090). Only 1 of 3 lung cell types (macrophages) produced individually significant geometric signal.

Table 5: **Representation repair through multi-layer bundling and seed ensembling.** All values are mean ΔAUROC across 3 seeds (42, 43, 44) on the external-lung domain unless otherwise noted. The gap column shows Geneformer minus scGPT. Final row shows seed-ensemble bundled results.

scGPT Representation	scGPT ΔAUROC	GF ΔAUROC	Gap (GF−scGPT)
External lung (mean across seeds)			
Single layer (L3)	+0.003	+0.025	+0.025
Bundle L1–4	+0.016	+0.025	+0.012
Bundle L0–5	+0.023	+0.025	+0.005
Bundle L0–11	+0.026	+0.025	+0.001
Seed-ensemble L0–11	+0.026	+0.024	−0.001
Cross-domain transfer (seed-ensemble L0–11)			
Immune	+0.066	+0.054	−0.013
Lung	+0.039	+0.028	−0.011
External lung	+0.026	+0.024	−0.001

Table 6: **Compact stacking model under leakage-resistant outer-split evaluation.** Baseline, scGPT, and Geneformer columns show single-branch AUROCs. The compact model combines both branches via stacked logistic regression with OOF predictions. Δ shows compact minus best single branch.

Domain	Baseline	scGPT	Geneformer	Compact	Δ
Immune	0.642	0.711	0.697	0.728	+0.017
Lung	0.571	0.612	0.600	0.622	+0.010
External lung	0.593	0.619	0.617	0.630	+0.011

Naive feature fusion was ineffective. Simply adding both models' geometric features to a single logistic regression (without stacking) produced gains of only +0.001 over Geneformer alone. The stacking architecture's advantage comes from learning a context-dependent combination policy.

Discussion

What geometric regulatory signal represents biologically

The central finding of this work is that the geometric arrangement of gene vectors in foundation model representation spaces encodes genuine information about transcription factor–target regulatory relationships. This signal is not reducible to expression-level confounds and is independently confirmed across two architectures trained on overlapping but distinct datasets.

We propose that this signal reflects *co-regulatory program structure*: genes that participate in shared or coupled regulatory cascades are encoded in related directions within the representation space, because the models learn to predict expression patterns shaped by regulatory relationships. This is consistent with the distributional hypothesis applied to transcriptomic data—genes that "co-occur" in similar expression contexts develop similar representations, and regulatory relationships generate specific co-occurrence patterns that leave geometric imprints.

This interpretation implies that geometric signal is a proxy for regulatory program membership, not a direct readout of TF binding. The signal should be strongest for regulatory relationships

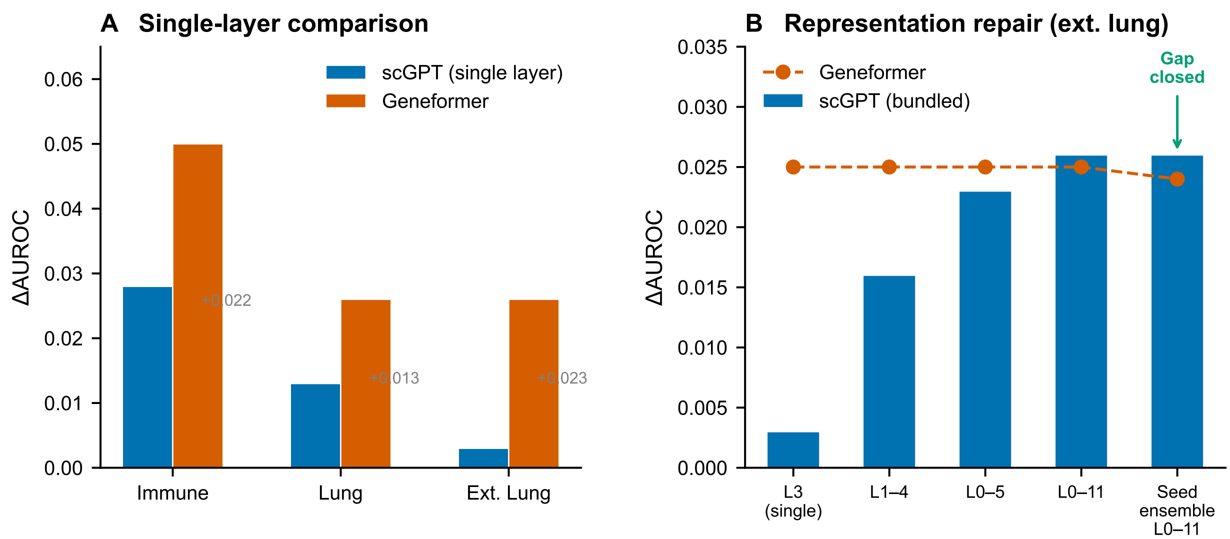

Figure 4: **Cross-model comparison and representation repair. (A)** Single-layer scGPT (blue) vs. Geneformer (vermillion) ΔAUROC across tissue domains on shared edges. Geneformer shows consistently larger signal, with the gap particularly pronounced in external lung. Grey annotations show the Geneformer–scGPT gap. **(B)** Representation repair in the external-lung domain through progressive multi-layer bundling of scGPT features. Bars show scGPT ΔAUROC as more layers are included; the dashed line shows the Geneformer reference. The gap closes monotonically and is eliminated by the full L0–11 seed-ensemble bundle.

that produce clear, consistent expression signatures, and weakest for context-specific or transient interactions. This prediction is consistent with our observation that the signal is strongest in CD8$^+$ T cells, where transcriptional reprogramming is dramatic and well-characterized, and weakest in relatively quiescent epithelial populations.

The distributed nature of regulatory information in transformers

Perhaps the most important methodological insight is that regulatory information in transformer residual streams is distributed across layers and stochastic across random initializations. The dramatic improvement from single-layer to multi-layer bundled scGPT representations—from near-null to competitive with Geneformer in the external-lung domain—demonstrates that different layers encode qualitatively different aspects of regulatory geometry.

This finding parallels results in NLP, where syntactic information concentrates in early-to-middle layers and semantic information in later layers of BERT-style models [19]. It suggests that interpretability studies of biological transformers should routinely evaluate multi-layer representations rather than selecting single "best" layers, which risks missing information encoded at non-optimal depths.

Complementarity between model architectures

The partial complementarity between scGPT and Geneformer signals has both scientific and practical implications. The moderate edge-level correlation ($\rho \approx 0.47$) quantifies this complementarity: enough overlap to suggest both models capture real biology, but enough divergence to enable

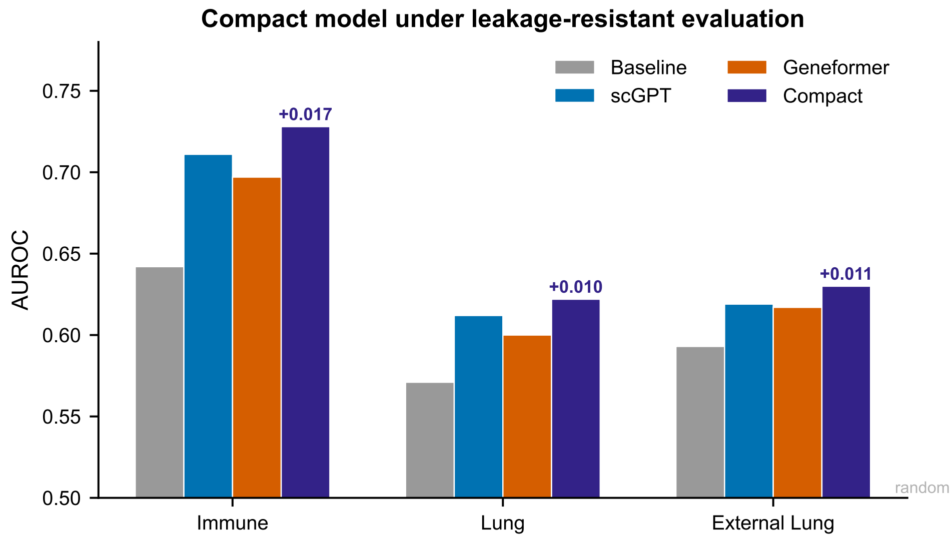

Figure 5: **Compact stacking model performance under leakage-resistant evaluation.** AUROC for baseline-only (grey), scGPT branch (blue), Geneformer branch (vermillion), and compact stacking model (indigo) across three tissue domains. The compact model consistently outperforms the best single branch, with gains annotated above each bar. All evaluations use nested outer-split cross-validation to prevent information leakage. The dotted grey line indicates random performance (AUROC = 0.5).

genuine information gain through combination. The enrichment of specific TF families in high-disagreement regions (GATA1, KLF1 in immune; HNF4A in lung) suggests that disagreement is biologically structured, reflecting genuine differences in how each model encodes specific regulatory programs.

Practically, this complementarity means that future improvements in either architecture can contribute additive value to the combined system. The compact stacking framework provides a principled mechanism for integrating such improvements.

Comparison with attention-based approaches

Our findings should be interpreted alongside recent work showing that attention patterns in single-cell transformers primarily capture co-expression rather than direct regulatory relationships [18]. Taken together, these results suggest that while attention weights reflect statistical co-expression, deeper residual-stream representations encode subtler regulatory structure. This distinction is consistent with the transformer architecture's design: attention computes pairwise token interactions within a layer, while the residual stream accumulates information across all layers through additive skip connections.

Implications for gene regulatory network inference

Our results suggest a concrete role for foundation model geometry in GRN inference pipelines. We envision three practical applications:

1. **Edge re-ranking:** Geometric scores can complement expression-based methods for re-ranking candidate regulatory edges.

2. **Hypothesis prioritization:** For experimental follow-up (e.g., CRISPRi screens), geometric scores provide orthogonal evidence beyond expression correlation.

3. **Context-dependent assessment:** Because foundation model representations are conditioned on cellular context, geometric features can capture tissue-specific regulatory relationships.

Limitations

Several limitations should be considered. First, TRRUST provides literature-curated edges that are biased toward well-studied genes and may contain false positives and false negatives. Second, effect sizes are modest in absolute terms, particularly in lung tissues (ΔAUROC $\approx +0.010$ to $+0.017$). Third, the analysis is retrospective; prospective validation on independent cohorts would strengthen the conclusions. Fourth, biological causality is not established—geometric proximity correlates with regulatory relationships but does not demonstrate mechanistic understanding. Fifth, computational cost is non-trivial, though extraction is a one-time cost per model and dataset.

Conclusions

We have presented a systematic investigation of whether residual-stream geometry in single-cell foundation models encodes gene regulatory information. Through analysis spanning four tissue contexts, two foundation model architectures, multiple geometric metrics, and comprehensive null controls, we establish three principal findings.

First, residual-stream geometry carries genuine incremental signal for regulatory edge classification beyond expression-level confounds, with effect sizes strongest in tissues and cell types with active transcriptional programs. Second, extraction of this signal is critically dependent on methodological choices—particularly centered-cosine similarity, multi-layer bundling, and seed ensembling—and initial negative results in some domains were largely methodological rather than biological. Third, two independently trained models encode partially complementary regulatory information that can be productively combined through stacked modeling, yielding consistent improvements across all tested domains under leakage-resistant evaluation.

These findings advance our understanding of what single-cell foundation models learn about gene regulation and provide a practical framework for leveraging that knowledge in GRN inference. As foundation models continue to grow in scale and sophistication, systematic geometric analysis of their representations offers a promising path toward more reliable computational predictions of gene regulatory architecture.

Methods

Data sources and preprocessing

Single-cell expression data. We used processed single-cell RNA-seq datasets from the Tabula Sapiens atlas [11], which provides comprehensive transcriptomic profiles across human tissues. Four tissue contexts were analyzed: **kidney** (used for initial signal detection), **immune** (a composite of immune cell types including B cells, $CD4^+$ T cells, and $CD8^+$ T cells), **lung** (containing alveolar type I, alveolar type II, and macrophage populations), and **external lung** (an independent lung cohort used for orthogonal validation).

Regulatory ground truth. As a reference set of regulatory interactions, we used the TRRUST v2 database [12], which curates experimentally validated TF–target relationships extracted from published literature. TRRUST provides binary regulatory edges (TF $\rightarrow$ target) without directionality of effect. After mapping to each tissue's gene universe, the resulting edge sets varied by domain: kidney (288 positive edges), immune ($\sim$1,768 total edges), lung ($\sim$15,384 edges), and external lung ($\sim$17,214 edges).

Negative edge construction. For each domain, we constructed matched negative edges by sampling random (source TF, target gene) pairs from the same source and target gene pools as the positive edges, with a 3:1 negative-to-positive ratio (kidney) or balanced 1:1 sampling (cross-domain experiments).

Foundation models

scGPT. We used the pre-trained scGPT whole-human model [8], a 12-layer transformer with generative pre-training on over 33 million single-cell profiles. For each tissue context, we sampled 256 cells (default seed 42, with additional seeds 43 and 44 for replication) and extracted per-layer residual-stream activations using forward hooks. The token budget was set to 512 genes per cell. Per-gene embeddings at each layer ℓ were obtained by averaging the corresponding token's residual-stream vector across sampled cells:

$$\mathbf{h}_g^{(\ell)} = \frac{1}{|C_g|} \sum_{c \in C_g} \mathbf{r}_{g,c}^{(\ell)}, \tag{1}$$

where $\mathbf{r}_{g,c}^{(\ell)} \in \mathbb{R}^d$ is the residual-stream vector for gene g in cell c at layer ℓ, C_g is the set of cells in which gene g appears in the token sequence, and d is the model's hidden dimension.

Geneformer. We used Geneformer [9] with its pre-trained token embeddings. Gene embeddings were extracted from the model's embedding layer, producing a single static vector per gene. Gene mapping between the Geneformer vocabulary and each tissue's expressed genes yielded coverage rates of 41.4% (immune), 83.2% (lung), and 77.2% (external lung).

Geometric feature extraction

For each candidate TF–target pair (g_s, g_t), we computed pairwise geometric features from the gene embeddings. Let $\mathbf{h}_s$ and $\mathbf{h}_t$ denote the source and target gene vectors, and let $\bar{\mathbf{h}} = \frac{1}{|G|} \sum_{g \in G} \mathbf{h}_g$ denote the global mean across all gene vectors. The following metrics were evaluated:

$$\text{Cosine similarity:} \quad s_{\cos}(g_s, g_t) = \frac{\mathbf{h}_s^\top \mathbf{h}_t}{\|\mathbf{h}_s\| \, \|\mathbf{h}_t\|}, \tag{2}$$

$$\text{Centered cosine:} \quad s_{\text{cc}}(g_s, g_t) = \frac{(\mathbf{h}_s - \bar{\mathbf{h}})^\top (\mathbf{h}_t - \bar{\mathbf{h}})}{\|\mathbf{h}_s - \bar{\mathbf{h}}\| \, \|\mathbf{h}_t - \bar{\mathbf{h}}\|}, \tag{3}$$

$$\text{Negative } L_2 \text{ distance:} \quad s_{L_2}(g_s, g_t) = -\|\mathbf{h}_s - \mathbf{h}_t\|_2, \tag{4}$$

$$\text{Dot product:} \quad s_{\text{dot}}(g_s, g_t) = \mathbf{h}_s^\top \mathbf{h}_t. \tag{5}$$

Low-rank variants. For PCA-projected features, we computed the top-k principal components of the gene embedding matrix $\mathbf{H} \in \mathbb{R}^{|G| \times d}$ for $k \in \{32, 64, 128\}$, projected all gene vectors into this subspace, and computed pairwise metrics in the reduced space.

Multi-layer bundling. We concatenated geometric features across multiple scGPT layers rather than selecting a single best layer. For a layer bundle $\mathcal{L} = \{0, 1, \dots, L\}$, the feature vector for a candidate edge becomes $\mathbf{f}_{g_s, g_t} = [s^{(\ell)}(g_s, g_t)]_{\ell \in \mathcal{L}}$.

Seed ensembling. We averaged geometric features across multiple random seeds (42, 43, 44) before downstream classification: $\bar{s}(g_s, g_t) = \frac{1}{3} \sum_i s^{(\mathrm{seed}_i)}(g_s, g_t)$.

Edge classification framework

For each tissue domain, we formulated a binary classification task: given a candidate TF–target pair, predict whether it is a curated TRRUST regulatory edge (positive) or a matched non-regulatory pair (negative).

Baseline features. All models included baseline confound features: mean expression, variance, and detection frequency for both source and target genes (six features total).

Model comparison. For each domain, we compared: (1) **Baseline-only**: logistic regression on six confound features; (2) **Baseline + geometry**: logistic regression adding geometric features to the baseline. The primary metric is ΔAUROC: the AUROC difference between the geometry-augmented and baseline-only models.

Evaluation protocol

All predictive evaluations used repeated stratified 5-fold cross-validation (3 repeats). For each ΔAUROC estimate, 95% bootstrap confidence intervals were computed using 1,000 resamples. For the compact stacking model, we used nested cross-validation: in the outer loop, data is split into train and test folds; within each outer fold, branch models are trained using inner cross-validation, and out-of-fold predictions serve as features for the meta-learner.

Null controls

Label permutation. We randomly permuted edge labels while preserving feature structure, then re-estimated ΔAUROC under permuted labels ($n = 40$ repeats). The empirical p-value is the fraction of permutation values exceeding the observed ΔAUROC.

Geometry-feature shuffle. We randomly permuted geometric scores across edges while preserving baseline features and labels ($n = 60$ repeats). This null tests whether the geometric feature identity, not merely the presence of an additional feature, drives the observed improvement.

Compact stacking model

The compact combiner integrates geometric signals from both foundation models through a stacked architecture: (1) train branch model A (baseline + scGPT) and B (baseline + Geneformer); (2) generate inner-fold OOF probability predictions; (3) train meta-model on baseline features + OOF predictions; (4) evaluate on held-out outer folds.

Declarations

Ethics approval and consent to participate

Not applicable. This study used only publicly available datasets (Tabula Sapiens atlas, TRRUST database) and did not involve human subjects, animal experiments, or clinical data requiring ethics approval.

Consent for publication

Not applicable.

Availability of data and materials

The Tabula Sapiens atlas is available at https://tabula-sapiens-portal.ds.czbiohub.org/. The TRRUST v2 database is available at https://www.grnpedia.org/trrust/. The scGPT pre-trained model is available at https://github.com/bowang-lab/scGPT. The Geneformer pre-trained model is available on Hugging Face at https://huggingface.co/ctheodoris/Geneformer. All analysis code, environment specifications, and reproduction instructions are available at https://github.com/Biodyn-AI/geometric-residual-stream.

Competing interests

The author declares no competing interests.

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Authors' contributions

IK conceived the study, designed and conducted all computational analyses, and wrote the manuscript.

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